UNIT-4

Test Planning and Automation

TESTING PLANNING

It is rightly sad that “ Failing to plan is planning to fail ”. Therefore testing also requires proper planning for appropriate execution.

Testing planning includes:

1. Preparing a Test Plan
2. Scope Management: deciding Features to be Tested/Not tested
3. Deciding Test Approach/Strategy
4. Setting up Criteria for Testing

5. Identifying Resources requirements

6. Identifying Test Deliverables

7. Testing Tasks : Size and Effort Estimation

8. Activity Breakdown and Scheduling

9. Communication Management

10. Risk Management

TESTING PLANNING

1. Preparing a Test Plan

Testing – like any project – should be driven by a plan. The test plan acts as the anchor for the execution, tracking, and reporting of the entire testing project and covers:

- What needs to be tested
- How the testing is going to be performed
- What resources are needed for testing
- The time lines by which the testing activities will be performed.
- Risks that may be faced in all of the above, with appropriate mitigation and contingency plans.

2. Scope Management : Deciding Features to be Tested/Not Tested

Scope management pertains to specifying the scope of a project. For testing, scope management entails

1. Understanding what constitutes a release of a project;
2. Breaking down the release into features;
3. Prioritizing the features for testing;
4. Deciding which features will be tested and which will not be; and
5. Gathering details to prepare for estimation of resources for testing.

The following factors drive the choice and prioritization of features to be tested.

- Features those are new and critical for the release.
- Features whose failures can be catastrophic.
- Features that are expected to be complex to test.
- Features which are extensions of earlier features that have been defect prone.

3. Deciding test approach/strategy

This includes identifying,

- What type of testing would you use for testing functionality?
- What are the configurations or scenarios for testing the feature?
- What integration testing would you do to ensure these features work together?
- What localization validations would be needed?
- What “non-functional” tests would you need to do?

4. Setting up Criteria for Testing

There must be clear entry and exit criteria for different phases of testing. There may also be entry criteria for the entries testing activity to start. The completion/exit criteria specify when a test cycle or a testing activity can be deemed complete.

5. Identifying Responsibilities, Staffing, and Training Needs

These different roles should complement each other. The different role definitions should

1. Ensure there is clear accountability for a given task, so that each person knows what he or she has to do;
2. Clearly list the responsibilities for various functions to various people, so that everyone knows how his or her work fits into the entire project;
3. Complement each other, ensuring no one steps on an others toes; and
4. Supplement each other, so that no task is left unassigned.

6. Identifying Resource Requirements

Some of the following factors need to be considered.

- Machine configuration (RAM, processor, disk, and so on) needed to run the product under test
- Overheads required by the test automation tool, if any
- Supporting tools such as compilers, test data generators, configuration management tools, and so on
- The different configurations of the supporting software (for example, OS) that must be present
- Special requirements for running machine-intensive tests such as load tests and performance test
- Appropriate number of licenses of all the software.

7. Identifying Test Deliverables

The deliverables includes the following, all reviewed and approved by the appropriate people.

1. The test plan itself (master test plan, and various other test plans for the project)
2. Test case design specifications
3. Test cases, including any automation that is specified in the plan.
4. Test logos produced by running the tests
5. Test summary reports.

8. Testing Tasks: Size and Effort Estimation

The scope identified above gives a broad overview of what needs to be tested. This understanding is quantified in the estimation step. Estimations happens broadly in three phases.

1. Size estimation
2. Effort estimation
3. Schedule estimation.

- **Size estimation:** It quantifies the actual amount of testing that needs to be done several factors contribute to the size estimate of testing project.

- **Size of the product under test**. This obviously determines the amount of testing that needs to be done. The larger the product, in general, grater is the size of testing to be done. Some of the measurement of the size of product under test are as follows;
  - Line of Code (LOC).
  - A function point (FP).
  - No of screens reports or transactions.

- **Extent of automation required** when automation is involved, the size of work to be done for testing increases.

- **Number of platforms and inter-operability environments to be tested**. If a particular product is to be tested under several different platforms or under several different configurations, then the size of the testing task increases.

Effort Estimation: Estimating effort is important because often effort has a more direct influence on cost than size. The other factors that drive the effort estimate are as follows:

- **Productivity data**:
- **Reuse opportunities** :
- **Robustness of process:**

For example;

1. Well-documented standards for writing test specifications, test scripts etc.
2. Proven process for performing functions such as reviews and audits;
3. Consistent ways of training people, and
4. Objective ways of measuring the effectiveness of compliance to process

9. Activity Breakdown and Scheduling

The following steps make up this translation.

1. Identifying external and internal dependencies among the activities.
2. Sequencing the activities, based on the expected duration as well as on the dependencies.
3. Monitoring the progress in terms of time and effort
4. Rebalancing schedules and resources as necessary.

External dependencies of an activity

Examples include

- Availability of the product from developers
- Hiring
- Training
- Acquisition of hardware/software required for training and
- Availability of translated message files for testing

Internal dependencies

Examples include

- Completing the test specification
- Coding/scripting the tests
- Executing the tests

Scheduling

10. Communications Management

Communications management consists of evolving and following procedures for communication that ensure that everyone is kept in sync with the right level of detail.

11. Risk Management

Risks are uncertain events that could potentially affect a project's outcome. Risk management entails

- Identifying the possible risks
- Quantifying the risks;
- Planning how to mitigate the risks; and
- Responding to risks when they become a reality

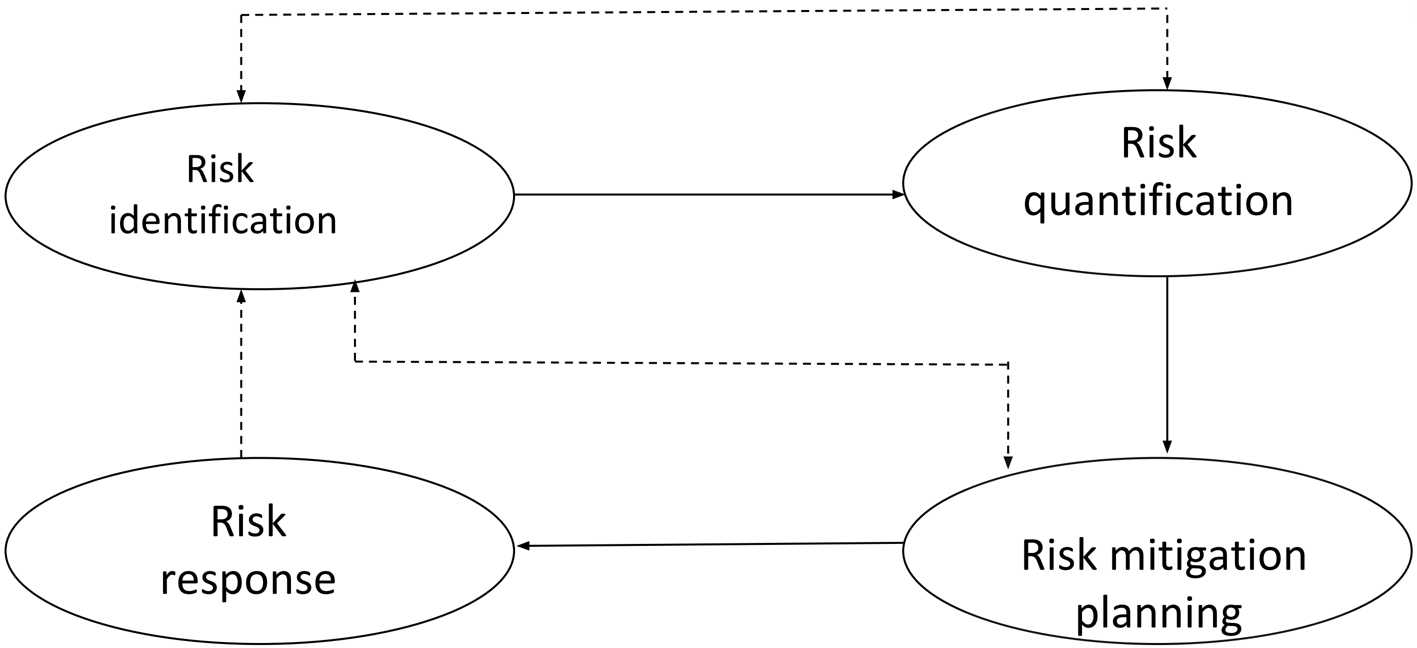

Figure: Aspects of risk Management

As some risks are identified and resolved, other risks may surface. Hence as risks can happen any time, risk management is essentially a cycle, which goes through the above four steps repeatedly.

I. ***Risk identification*** consists of identifying the possible risks that may hit a project. The following are some of the common ways to identify risks in testing.

1. Use of checklists
2. Use of organizational history and metrics
3. Informal networking across the industry

II. ***Risk quantification*** deals with expressing the risk in numerical terms. There are two components to the quantification of risk. One is the probability of the risk happening and the other is the impact of the risk, if risk happens.

Ex: The occurrence of a low priority defect may have a high probability but a low impact.

III. Risk mitigation planning deals with identifying alternative strategies to combat a risk event, should that risk materialize

Ex: A couple of mitigation strategies for the risk of attrition are to spread the knowledge to multiple people and to introduce organization-wide processes and standards.

When the above three steps are carried out systematically and in a timely manner, the organization would be in a better position to respond to the risks, should the risks become a reality.

The following are some of the common risks encountered in testing projects and their characteristics.

- Unclear requirements
- Schedule dependence

- Insufficient time for testing
- “Show stopper” defects.
- Availability of skilled and motivated people for testing.
- Inability to get a test automation tool.

IV. Risk response, here in the risk response, the project manager needs to choose strategies that will reduce the risk to minimal. Project managers can choose between the following four risk response strategies:

→ Avoidance

Eliminate the risk, determinate this risk not become involved

→ Reduction

Take correct measures and mitigate plan to reduce the impact of risks

→ Sharing

Transfer the risk to other resource such as insource or insure

→ Accept

Acknowledge the risk and prepare the budget for this risk

Software Test Automation

Test automation is the process of using automation tools to maintain test data, execute tests, and analyze test results to improve software quality.

Automation testing is a Software testing technique to test and compare the actual outcome with the expected outcome. This can be achieved by writing test scripts or using any automation testing tool. Test automation is used to automate repetitive tasks and other testing tasks which are difficult to perform manually.

Automation saves the time as software can execute test cases faster than humans do, The time thus saved can be used effectively for test engineers to,

- Develop additional test cases to achieve better coverage
- Perform some esoteric or specialized tests like ad hoc testing
- Perform some extra manual testing

What are the types of Automation Testing?

There are certain tests that should be automated using test automation tools. Some of them are as follows:

- Unit Testing
- Integration Testing
- Regression testing
- Performance testing
- Security testing

Advantages of Automation Testing

- The time saved in automation can also be utilized to develop additional test cases, thereby improving the coverage of testing.
- Test automation can free the test engineers from mundane tasks and make them focus on more creative tasks
- Automated tests can be more reliable
- Automation helps in immediate testing
- Test automation opens up opportunities for better utilization of global resources

Which Test Cases to Automate

It is impractical to automate all testing, so it is important to determine what test cases should be automated first.

You can get the most benefit out of your automated testing efforts by automating:

- Repetitive tests that run for multiple builds.
- Tests that tend to cause human error.
- Tests that require multiple data sets.

- Frequently used functionality that introduces high risk conditions.
- Tests that are impossible to perform manually.
- Tests that run on several different hardware or software platforms and configurations.
- Tests that take a lot of effort and time when manual testing.

Which test cases should you not automate

- Subjective Validation
- New Functionalities
- Business-critical Functionalities
- User Experience
- Complex Functionalities
- Quality Control

Manual Testing Vs Automation Testing

Manual Testing	Automation Testing
Test cases are executed manually.	Test cases are executed with the help of tools.
Reliability is less.	Reliability is more.
It is less costlier.	It is more costlier.
For some test cases it consumes time.	As it is a machine it take less time to execute cases.
Human can make mistakes and hence accuracy is less.	Machine hardly makes mistakes(If it has been asked to do so).
As it include human intervention, it is beneficial to check ease for accessing the application.	It includes tools so unable to check usability or accessibility.

Design and Architecture for Automation

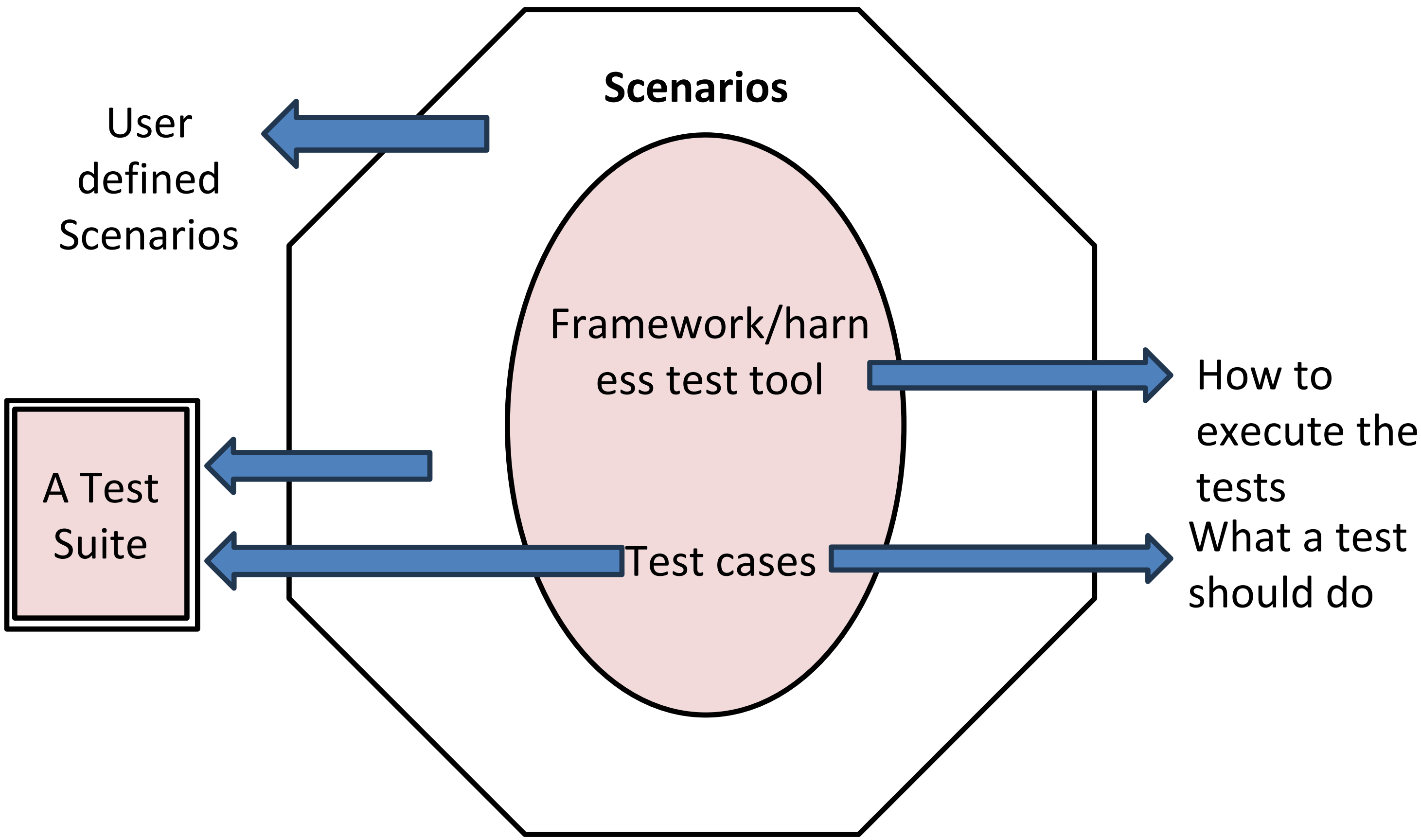

Figure: Framework for test automation

Scope of Automation

The first phase involved in product development is requirements gathering it is no different for test automation as the output of automation can also be considered as a product. Some generic tips for identifying the scope for automation are:

- Identifying the types of Testing Amenable to Automation
  - Stress, reliability, scalability and performance testing
  - Regression tests
  - Functional tests
- Automating areas less prone to change
- Automate tests that pertain to standards
- Management aspects in Automation

Process of Test Automation

1. Convince the Management
2. Finding Automation tool experts
3. Using the correct tool for automation
4. Training the Team
5. Creating the test automation Framework
6. Developing an Execution Plan
7. Writing Scripts
8. Reporting
9. Maintenance of Scripts

Thank You